Rachit Tamrakar

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LinkedIn | Portfolio

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Profile

Mechatronics student with a strong background in building and integrating embedded systems, electronics, and control software. Developed reliable hardware-software solutions for engineering projects involving real-time control, testing, and data acquisition. Quick to learn new tools and technologies, and comfortable working across technical domains in fast-paced, collaborative environments.

Work Experience

Bosch Australia Manufacturing Solutions (BAMS) — Melbourne, Australia

Undergraduate Vision, Robotics and Simulations Engineer — *Mar 2024 - Present*

Vision Inspection System

- Developed a multi-station vision inspection system for real-time validation of products across an **$8M** production line (team of 2).
- Designed and implemented a **C#** application handling **multithreading**, camera integration, **PLC** communication, **SQL** logging, and custom UI.

Real-Time Object Detection System using YOLO

- Created a real-time object detection pipeline for accurate product positioning on a live manufacturing line.
- Built dataset automation tools for image collection, labeling, and model training, training a compact **YOLO** model and improving inference time by **1.5x**.
- Met strict speed requirements using transfer learning on a small **YOLO** backbone.

Digital Twin Simulation in NX Mechatronics Concept Designer

- Developed a digital twin to test new robotic palletizer features without halting production (team of 3).
- Built high-fidelity **NX** models and automated simulation setup with Python scripting, reducing test preparation time by **2x**.

Projects / Extracurriculars

Monash High Powered Rocketry (Monash HPR) — Melbourne, Australia

Avionics Vice Lead / Engineer — *Aug 2023 - Present*

Solaris Mk II Hybrid Engine & Ground Station Electronics (IREC 2025)

- Flown on Project Zenith at IREC 2025, resulting in Monash HPR winning the Jim Furfaro Award for Technical Excellence (out of **140+** teams) and placing **2nd** in category.
- Worked in a team of 5 to develop the electronics and software systems supporting a **4kN** hybrid engine and ground station, enabling full real-time control, monitoring, and data logging.
- Personally owned the ground station project, covering design, fabrication, testing, and integration with the engine control architecture.

Capacitive Propellant Sensor

- Led a 4-person team to develop a capacitive N2O propellant-level sensor for accurate in-tank mass measurement and safer engine fueling.
- Designed the sensing probe, wrote firmware, and performed calibration to deliver repeatable measurements with accuracy within +/**- 10 g**.
- Resolved electrical noise and stability issues through shielding, probe design tweaks, and calibration procedures.

Education

Monash University — Melbourne, Australia

Bachelor of Engineering - Mechatronics (Honours) — *Feb 2022 - Nov 2026*

High Distinction (83.714 WAM, 3.64 GPA) | Deans Honours List (2022, 2023)

Skills

General: Systems Engineering, Computer Networks, 3D Printing, Algorithms & OOP, DAQ & Instrumentation Systems
Programming: Git, C/C++, C#, .NET, Python, **Multithreaded Programming**, Embedded Firmware, Network Programming (TCP/UDP, MQTT, RS422, CAN), SQL, MATLAB, Simulink, PyTorch, OpenCV, Signal Processing
Automation & Design: Siemens PLC, SCADA, NX, SolidWorks, Creo Parametric, Altium Designer, STM32CubeIDE
Platforms: Linux, Raspberry Pi, VMware, STM32, Arduino, Siemens PLC, Adobe CC, MS Office